Trees with perfect matchings that disprove a conjecture of Ramos and Sun

Heshun Fan
School of Mathematics and Statistics
Huazhong University of Science and Technology
Wuhan 430074, China
U202410055@hust.edu.cn

Abstract

We construct an explicit family of trees with perfect matchings whose independence sequences fail log-concavity one position below the independence number. For every integer $p \geq 6$, the tree has $14p + 6$ vertices and independence number $m = 7p + 3$, while its independent-set counts satisfy $a_{m-1}^2 < a_{m-2}a_m$. This disproves Conjecture 4.2 of Ramos and Sun, which predicts a lower bound on the independence number from the position of a failure of log-concavity. The proof encodes independent sets by endpoint choices on matching pairs and compares counts obtained by deleting pair labels. It also gives $a_{m-1}^2/(a_{m-2}a_m) < 49 \cdot 2^{-p}$. We give exact coefficients for the 90-vertex instance and describe its Lean formalization.

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1 Introduction

For a finite graph G, let $a_k(G)$ denote the number of independent vertex sets of cardinality k, and let $\alpha(G)$ denote its independence number. The independence polynomial is

$$I_G(x) = \sum_{k=0}^{\alpha(G)} a_k(G)x^k.$$

The sequence is log-concave at an interior index $1 \leq j < \alpha(G)$ if $a_j(G)^2 \geq a_{j-1}(G)a_{j+1}(G)$. All coefficients from $a_0(G)$ to $a_{\alpha(G)}(G)$ are positive, since subsets of a maximum independent set realize every such size. Consequently, log-concavity implies unimodality. These two shape properties have motivated the study of independence polynomials of trees; see the background in Ramos and Sun [8, Section 2.1]. Trees with non-log-concave independence sequences are already known [7]. A more specific question asks how the position of a failure constrains the independence number.

Put $h = \lfloor N/2 \rfloor$ for a tree T of order N. Motivated by their computational searches, Ramos and Sun proposed the following implication in Conjecture 4.2 of [8]:

$$a_{h-\beta}(T)^2 < a_{h-\beta-1}(T)a_{h-\beta+1}(T) \quad \Longrightarrow \quad \alpha(T) \geq h + \beta + 1. \tag{1.1}$$

Their convention in Section 1.2 replaces $N/2$ by its floor when N is odd. The conjecture does not separately specify a range for β. We disprove the even-order case with $\beta = 1$ and with all three coefficient indices in the positive support. In this case, failure at $N/2 - 1$ would require $\alpha(T) \geq N/2 + 2$.

Every tree of order N has independence number at least $\lceil N/2 \rceil$, by its bipartition. For a tree of order $2m$ with a perfect matching, each independent set uses at most one endpoint of each

matching edge, while each bipartition class has size m. Such a tree therefore has independence number exactly m. Our construction attains this value and fails log-concavity at $m-1$. Write $\nu(G)$ for the maximum size of a matching in G.

Theorem 1.1. *For each integer $p \geq 6$, construct T_p from six distinct vertices a, b, c, d, e, f with edges*

$$ab, \quad cd, \quad ef, \quad af, \quad cf$$

by attaching p pendant paths of length two at each of a, c, e, and $2p$ such paths at each of b, d. Each attached path has two new vertices, distinct from those of every other attached path. Then T_p is a tree with

$$|V(T_p)| = 14p + 6, \qquad \nu(T_p) = \alpha(T_p) = 7p + 3.$$

Writing $m = 7p + 3$ and $a_k = a_k(T_p)$, we have

$$a_{m-1}^2 < a_{m-2} a_m. \tag{1.2}$$

Thus T_p contradicts (1.1) *with $\beta = 1$.*

Indeed, the conjecture would require $\alpha(T_p) \geq m + 2$, whereas $\alpha(T_p) = m$. The proof also shows that $a_{m-1}^2/(a_{m-2}a_m) < 49 \cdot 2^{-p}$, so this ratio tends to zero along the family; see Corollary 2.2. For $p = 6$, we obtain a 90-vertex counterexample with independence number 45 and failure at index 44.

The matching has a second role beyond fixing the independence number. An independent set leaves some matching pairs empty and selects one endpoint of each remaining pair. Thus the three highest-degree coefficients count configurations with zero, one, and two empty pairs. Attaching paths of length two preserves the matching, and the unequal numbers p and $2p$ of attached paths give different weights to the two endpoint choices in a core pair. Leaving two specified core pairs empty removes the restrictions on many pendant pairs at once. This gives a lower bound for a_{m-2} that, when multiplied by a_m, exceeds the square of the sum of all one-empty-pair contributions. Only a few exact counts are needed; the remaining contributions admit a common bound.

1.1 Relation to earlier constructions

To compare an established failure at index j with the proposed bound (1.1), set $\beta = h - j$. The proposed conclusion then reads

$$\alpha + j \geq 2\lfloor N/2 \rfloor + 1. \tag{1.3}$$

The comparisons below concern the specific failure indices established in the cited works. When $j > h$, this substitution gives a negative β rather than an instance with nonnegative β.

Kadrawi and Levit [7, Sections 3–4] give two types of infinite families extending their 26-vertex examples. The stated failures have $\alpha = N/2+1$ and $j = N/2$, so (1.3) holds with equality. Galvin [6, Theorem 2.1] constructs trees $T_{q,t,1}$ with

$$N = 1 + q + 2qt, \qquad \alpha = q(t+1), \qquad j = qt + 2$$

for integers $t \leq q \leq 2^{t/16}$ and sufficiently large t. Here $\alpha + j = N + 1$, which implies (1.3). Both types of Kadrawi–Levit trees and Galvin's trees have $\alpha > N/2$, in contrast to Theorem 1.1.

Bautista-Ramos [2, Definition 1 and Theorem 1] constructs trees with arbitrarily many failures. Denoting the number of branches by q, the parameters are

$$N = 2 + q(4 + 6t), \qquad \alpha = 1 + 3q(t+1), \qquad j = \alpha - (2r+1),$$

where $q \geq 1$ is a fixed integer, t is a sufficiently large integer, and $r \in \{0, \ldots, q-1\}$. In this case, $\alpha + j - N - 1 = 2(q - r - 1) \geq 0$, and $\beta = 1 - q + 2r$ is nonnegative precisely when $2r \geq$

$q-1$. Bautista-Ramos, Guillén-Galván, and Gómez-Salgado [3] develop rooted constructions and linear recurrences with consecutive failures. Their Corollary 4 gives $(N, \alpha, j) = (19q + 6, 10q + 3, 10q + 2)$ for integers $q \geq 9$. These parameters satisfy (1.3), since $\alpha + j - N - 1 = q - 2 > 0$, but give $\beta = 1 - \lceil q/2 \rceil < 0$. Galvin [6, Section 3] also reports examples computed by Bencs with several failures, without specifying their indices; those data do not determine the comparison in (1.3).

In a September 2026 preprint, Fang, Lu, Nevo, Yao, and Zheng [5, Theorems 1.1–1.2] announce unimodality for sufficiently large forests and strict log-concavity in the range $N/5 \leq j \leq 17\alpha/25$. Our index $j = \alpha - 1$, with $\alpha \geq 45$, lies beyond that range. The local failure in Theorem 1.1 does not imply a failure of unimodality.

The representation of a tree by matching pairs and oriented edges also occurs in the public implementation [4]. Public Lean source [1] contains a correctness theorem linking rooted-tree polynomial calculations to independent-set counts, together with non-log-concave witnesses of orders 26, 28, and 29. The two even-order witnesses have $(N, \alpha, j) = (26, 14, 13)$ and $(28, 15, 14)$, and hence satisfy (1.3); the odd-order witness has $j = 15 > \lfloor 29/2 \rfloor$. The formalization described here concerns the 90-vertex instance of Theorem 1.1.

Organization. Section 2 proves Theorem 1.1 by counting assignments on matching pairs and derives the bound on the coefficient ratio. Section 3 gives the exact 90-vertex instance and the scope of its formal verification. Section 4 discusses the parameter threshold and further coefficient estimates. Appendix A records an independent polynomial recurrence and the formal source identifiers.

Conventions and notation. All graphs are finite and simple. The length of a path is its number of edges. Vertex deletion is induced deletion, and $N_G[v]$ denotes the closed neighborhood of v. We set $a_k(G) = 0$ for $k > \alpha(G)$ and $I_\varnothing(x) = 1$.

2 Counting independent sets through a perfect matching

We prove Theorem 1.1 by using the perfect matching to count independent sets. The matching first determines the independence number; the resulting representation of independent sets then reduces the coefficient comparison to counts on a directed tree of matching pairs.

2.1 The tree and its matching pairs

Fix an integer $p \geq 6$, put $m = 7p + 3$, and abbreviate $a_k = a_k(T_p)$ for the rest of the proof. The six-vertex core is connected with five edges, hence is a tree. Attaching a path with two new vertices preserves connectedness and acyclicity. There are $7p$ attached paths, so the resulting tree has $6 + 14p = 2m$ vertices. The three core edges ab, cd, ef, together with the outer edge of every attached path, form a perfect matching of size m; see Figure 1. Thus $\nu(T_p) = m$.

Label the endpoints of the core matching edges by

$$(a, b) = (U_u, V_u), \quad (c, d) = (U_v, V_v), \quad (e, f) = (U_z, V_z).$$

Extend the bipartition along every attached path, using U and V for its two classes. Each matching pair is then $\{U_i, V_i\}$ for some label i. The m vertices U_i form an independent set. Conversely, any independent set contains at most one endpoint of each of the m matching edges. This proves $\alpha(T_p) = m$ directly.

Record every nonmatching edge U_iV_k as an arrow $i \to k$ in a directed tree $H = H_p$ on the matching-pair labels. Its internal arrows are $u \to z$ and $v \to z$. Each of u, v has p outgoing leaves and $2p$ incoming leaves, and z has p outgoing leaves. These $7p$ leaves are all distinct. Conversely, replacing each vertex i of this directed tree by the edge U_iV_i, and each arrow $i \to k$ by the edge U_iV_k, recovers exactly T_p.

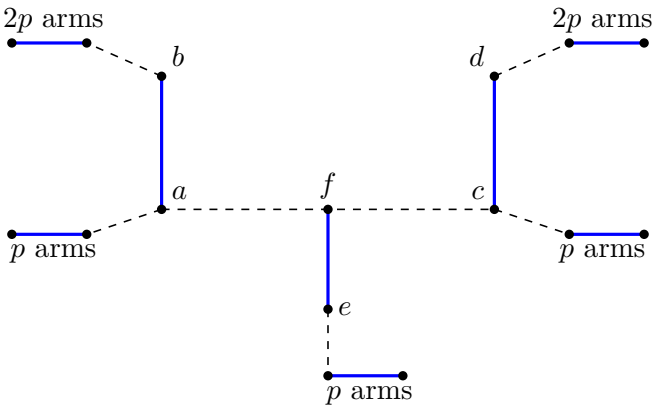


Figure 1: The six-vertex core and one representative of each group of pendant paths. Each displayed arm represents the stated number of distinct copies. Thick solid blue edges form the perfect matching; thin dashed black edges are nonmatching. No arm is attached at f.

2.2 Independent sets as assignments on pairs

We encode the choice of U_i by $+$ and the choice of V_i by $-$. An unused matching pair is represented by deleting its label from H. The arrows then record precisely which endpoint choices are incompatible.

For a finite directed graph K, let $\mathcal{C}(K)$ be the set of assignments $\sigma : V(K) \to \{+,-\}$ for which no arrow $i \to k$ has $(\sigma(i),\sigma(k)) = (+,-)$. Write $\Omega(K) = |\mathcal{C}(K)|$, including $\Omega(\varnothing) = 1$. Deleting vertices always means induced deletion: an arrow is retained exactly when both of its endpoints remain. In particular, no constraint is retained through a deleted vertex.

Grouping independent sets by their empty pairs gives the following identity. The leaf bound will control all pendant pairs at once.

Lemma 2.1. *For every integer $0 \le r \le m$,*

$$a_{m-r} = \sum_{\substack{Z \subseteq V(H) \\ |Z|=r}} \Omega(H - Z). \tag{2.1}$$

Moreover, for each leaf ℓ of H, $\Omega(H-\ell) \le \Omega(H)$.

Proof. An independent set of T_p chooses zero or one endpoint from each matching pair. Let Z consist of the empty pairs. On a nonempty pair, assign $+$ when U_i is chosen and $-$ when V_i is chosen. The nonmatching edge U_iV_k forbids exactly the choice $(+,-)$ on its arrow $i \to k$. Thus the choices on the nonempty pairs give an element of $\mathcal{C}(H-Z)$, and the independent set has size $m - |Z|$.

Conversely, given Z and an element of $\mathcal{C}(H-Z)$, choose U_i at every $+$, choose V_i at every $-$, and choose neither endpoint of each pair in Z. No matching edge has both endpoints chosen. For a nonmatching edge, either an endpoint pair is empty or the forbidden sign pattern is excluded by the assignment. The chosen set is therefore independent. These two constructions are inverse, which proves (2.1).

For the last assertion, restriction gives a surjection from $\mathcal{C}(H)$ onto $\mathcal{C}(H-\ell)$. Indeed, let t be the neighbor of ℓ. If the arrow is $t \to \ell$, a sign $+$ at t forces $+$ at ℓ, while a sign $-$ at t permits either sign. If the arrow is $\ell \to t$, a sign $-$ at t forces $-$ at ℓ, while a sign $+$ at t permits either sign. Every assignment therefore has at least one extension, and there are no other constraints involving the leaf. □

Put

$$w_0 = \Omega(H), \qquad w_x = \Omega(H-x), \qquad w_{xy} = \Omega(H-\{x,y\}) \quad (x \ne y).$$

In this notation, (2.1) gives

$$a_m = w_0, \qquad a_{m-1} = \sum_{x \in V(H)} w_x, \qquad a_{m-2} = \sum_{\{x,y\} \subseteq V(H)} w_{xy}. \tag{2.2}$$

The last sum is over unordered two-element subsets.

For a small illustration, take the path with vertices in the order V_1, U_1, V_2, U_2 and matching edges U_1V_1, U_2V_2. Its pair-label graph has the single arrow $1 \to 2$. With no empty pair the allowed assignments are $(+,+), (-,+), (-,-)$; with exactly one empty pair there are two choices for either empty label; with both pairs empty there is one choice. These counts give $I_{P_4}(x) = 1 + 4x + 3x^2$.

2.3 The coefficient comparison

Set $B = 2^p$. Incorporating the choices at the pendant leaves gives weights B^2 and B for signs $+$ and $-$ at either u or v, respectively. For example, with sign $+$ at u, its $2p$ incoming leaves are free and its p outgoing leaves are forced to $+$. With sign $-$ at u, the incoming leaves are forced to $-$ and the outgoing leaves are free. At z the weights are 1 and B for signs $+$ and $-$.

If z has sign $+$, both signs are permitted at u, v; if z has sign $-$, both u, v must have sign $-$. Consequently

$$w_0 = (B^2 + B)^2 + B^3 = B^4 + 3B^3 + B^2. \tag{2.3}$$

Deleting u makes its $3p$ leaves isolated, with B^3 assignments. For the remaining v, z system, sign $+$ at z contributes $B^2 + B$ and sign $-$ contributes B^2. The interchange of u and v is an automorphism preserving all arrows, so

$$w_u = w_v = B^3(2B^2 + B) = B^4(2B + 1). \tag{2.4}$$

Deleting z instead frees its p leaves and separates the other two branches. Deleting both u, v frees their $6p$ leaves and leaves just the z branch. Thus

$$w_z = B(B^2 + B)^2 = B^3(B + 1)^2, \qquad w_{uv} = B^6(B + 1). \tag{2.5}$$

For $B \geq 4$, one has $w_0 \leq 2B^4$, since $3/B + 1/B^2 \leq 13/16 < 1$. Each of the $7p$ leaf deletions has count at most w_0 by Lemma 2.1. Summing the three internal deletions and the $7p$ leaf deletions gives

$$\begin{aligned} a_{m-1} &\leq 2B^4(2B + 1) + B^3(B + 1)^2 + 14pB^4 \\ &= 5B^5 + (4 + 14p)B^4 + B^3. \end{aligned} \tag{2.6}$$

For every integer $p \geq 6$,

$$4 + 14p \leq \frac{3}{2} 2^p. \tag{2.7}$$

The base case is $88 \leq 96$. If the inequality holds for $p \geq 6$, then $4 + 14(p + 1) \leq 2(4 + 14p)$, because this last comparison reduces to $10 \leq 14p$. This proves the induction step for (2.7). Since $B \geq 64$, (2.6) now yields

$$a_{m-1} \leq B^5 \left(\frac{13}{2} + B^{-2} \right) < 7B^5. \tag{2.8}$$

On the other hand, the nonnegative sum in (2.2) for a_{m-2} includes the summand w_{uv}. Equations (2.3) and (2.5) therefore imply

$$a_m \geq B^4, \qquad a_{m-2} \geq B^6(B + 1) \geq B^7. \tag{2.9}$$

As $49 < B$, the desired strict inequality follows:

$$a_{m-1}^2 < 49B^{10} < B^{11} \leq a_{m-2}a_m.$$

This proves (1.2) for every integer $p \geq 6$.

To finish the proof of Theorem 1.1, take $N = 2m$, $h = m$, $j = m - 1$ and $\beta = 1$. Then $j + \beta = h$, whereas $\alpha(T_p) = m < m + 2 = h + \beta + 1$. All three coefficients are positive: the independent set $\{U_i : i \in V(H)\}$ of size m has subsets of every cardinality from 0 to m. Thus the strict failure is at a genuine interior index and contradicts (1.1).

2.4 The ratio of the three highest-degree coefficients

The estimates used in the proof also quantify the failure of log-concavity. In particular, the ratio in the next corollary can be made arbitrarily small within the class of trees with perfect matchings.

Corollary 2.2. *For every integer $p \geq 6$, let $m = 7p + 3$ and $a_k = a_k(T_p)$. Then*

$$0 < \rho_p := \frac{a_{m-1}^2}{a_{m-2}a_m} < 49 \cdot 2^{-p}. \tag{2.10}$$

In particular, $\rho_p \to 0$ as $p \to \infty$ through the integers.

Proof. The coefficients are positive because $\alpha(T_p) = m$. With $B = 2^p$, equations (2.8) and (2.9) give

$$0 < \rho_p < \frac{(7B^5)^2}{B^7 B^4} = \frac{49}{B} = 49 \cdot 2^{-p}.$$

The upper bound tends to zero. □

3 The 90-vertex counterexample

For $p = 6$, Theorem 1.1 gives $N = 90$, $m = 45$, and $j = 44$. In this section write $a_k = a_k(T_6)$. We compute its three highest-degree coefficients explicitly and then describe the formal verification of this fixed instance.

3.1 Exact coefficients

Allow each matching pair to be empty, to select its U endpoint, or to select its V endpoint, and give each selected endpoint weight x. Put

$$\begin{aligned} A &= 1 + 2x, \qquad C = 1 + x, \\ W &= A^{18} + xA^{12}C^6 + xA^6C^{12}, \\ R &= A^{18} + xA^6C^{12}. \end{aligned} \tag{3.1}$$

A free leaf pair contributes A, whereas a leaf pair with one endpoint forbidden contributes C. For either u or v, the three terms of W correspond, in order, to the empty, $+$, and $-$ states. The polynomial R excludes the $+$ state. If z is empty or has sign $+$, it imposes no restriction on u, v; these two cases contribute A^6W^2 and xC^6W^2. If z has sign $-$, the $+$ state is forbidden at both u and v, giving xA^6R^2. Hence

$$I_{T_6}(x) = (A^6 + xC^6)W^2 + xA^6R^2. \tag{3.2}$$

Extracting the coefficients in (3.2) gives

$$\begin{aligned} a_{43} &= 4\,761\,097\,500\,672, \\ a_{44} &= 5\,965\,496\,320, \\ a_{45} &= 17\,567\,744, \end{aligned} \tag{3.3}$$

and hence

$$a_{44}^2 - a_{43}a_{45} = -48\,054\,595\,706\,911\,981\,568 < 0. \tag{3.4}$$

The independence number is 45 by Theorem 1.1, while the conjecture at $N = 90$, $h = 45$, and $\beta = 1$ requires at least 47.

3.2 Formal verification of the fixed instance

The fixed graph has been formalized in Lean as `graph90` of type `SimpleGraph (Fin 90)`. The formalization proves that this graph is a tree with independence number 45. Its counting argument establishes a size-preserving bijection between independent vertex sets and legal assignments of states $0, +, -$ to the matching pairs. The state 0 means that a pair is empty, and the other states select its U or V endpoint. Separating the three core pairs from the 42 pendant pairs reduces the polynomial calculation to the 27 core states.

The theorem `graph90_coefficient_eq_polynomial` proves that, for every nonnegative integer k, the coefficient of x^k in the right-hand side of (3.2) equals the number of independent vertex sets of size k in `graph90`. The final results establish (3.3), treehood, independence number 45, and the negation of the 90-vertex, $\beta = 1$ instance of (1.1). The all-parameter estimates in Section 2.3 are proved combinatorially here; the formal development covers $p = 6$. Appendix A.2 specifies its source files and verification records.

4 Further questions

Theorem 1.1 shows that the proposed bound fails even for trees with perfect matchings and independence number exactly half their order. The proof isolates a useful feature of this construction: the single contribution w_{uv} suffices for the lower bound on a_{m-2}, whereas every one-pair deletion must be controlled to bound a_{m-1}. This suggests examining other oriented trees of matching pairs for which the same imbalance of deletion counts can be established.

The assumption $p \geq 6$ ensures (2.7) and $2^p > 49$. It is a sufficient threshold for the estimates used here. Determining the least parameter for which this particular family violates (1.1) requires more precise coefficient calculations. This is distinct from determining the smallest order of a counterexample among all trees; no minimality assertion is made for the 90-vertex instance.

Corollary 2.2 gives an exponentially decreasing upper bound, but does not determine the exact asymptotic behavior of ρ_p. To obtain sharper estimates, one can account for the other two-pair deletion terms in (2.2) and refine the sum of the one-pair deletion counts. Such estimates may also help determine which choices of pendant-path multiplicities yield smaller examples.

A Independent coefficient calculations and formal records

The matching-pair calculation can be checked by recurrences applied directly to the tree. We record these recurrences first, followed by the identifiers of the formal statements for T_6.

A.1 Recurrences on the original tree

For any graph G and vertex v, partitioning independent sets according to whether they contain v gives

$$I_G(x) = I_{G-v}(x) + xI_{G-N_G[v]}(x). \tag{A.1}$$

Independent choices in distinct components give the product rule. A recorded independent Python calculation used these rules, with $I_\varnothing(x) = 1$, to recover (3.3) from the fixed edge list of T_6.

For a tree, a recurrence using a chosen root avoids repeated vertex deletions. Let S be rooted at o, let $\mathrm{Ch}(v)$ be the set of children of v, and let S_v be the subtree induced by v and its descendants. Denote by $F_v^{(0)}(x)$ and $F_v^{(1)}(x)$ the generating polynomials for independent sets of S_v excluding and including v, respectively. Then

$$\begin{aligned} F_v^{(0)}(x) &= \prod_{t\in \mathrm{Ch}(v)} \left(F_t^{(0)}(x) + F_t^{(1)}(x)\right), \\ F_v^{(1)}(x) &= x \prod_{t\in \mathrm{Ch}(v)} F_t^{(0)}(x). \end{aligned} \tag{A.2}$$

Empty products are 1, so the initial values at a leaf are $F_v^{(0)}(x) = 1$ and $F_v^{(1)}(x) = x$. At the root, $I_S(x) = F_o^{(0)}(x) + F_o^{(1)}(x)$. Indeed, excluding v allows either state at each child, whereas including v excludes every child. Distinct child subtrees have no edges between them, so their choices multiply; the factor x records the inclusion of v. All these calculations use integer polynomial arithmetic.

A.2 Formal source and recorded checks

The author retains the formal source and verification records locally in the unpublished artifact `research_003_rs42_lean90_complete_20261004`; these files have not been deposited in a public archive. The following identifiers refer to statements and files within that artifact. It contains the edge list `graph/tree90_edges.json` and the final source file `RS42Counterexample90Final.lean` under `lean/FormalProof/Research003/`. The theorem `graph90_adj_iff_fixedEdges` identifies `graph90` with that 89-edge list. The final source file contains the following statements:

- `graph90_actual_coefficients` proves the three equalities in (3.3).
- `graph90_counterexample` combines treehood, independence number 45, failure of log-concavity at 44, and failure of the required independence-number bound 47.
- `exists_counterexample90` states the corresponding existence theorem using `Nat.card` of the finite vertex sets satisfying `T.IsNIndepSet k s`, which means that s is independent in T and has cardinality k.
- `not_RamosSun42EvenBetaOne90` negates the universal 90-vertex, $\beta = 1$ instance of (1.1).

The final statements have no unproved treehood or coefficient hypotheses.

The artifact also contains `PROOF_MAP.md`, the records `verification/BUILD_REPORT.md` and `verification/FINAL_AXIOMS.txt`, and the independent Python checker and its output under `audit/historical_reproduction/`. The recorded Linux builds use Lean 4.34.0 and mathlib revision `56a9a88447f391672d7aeea117da03b6f7016b0d`, rebuilding all 12 local modules in fresh output directories with locked standard dependencies. The reported axiom dependencies of the final results are `propext`, `Classical.choice`, and `Quot.sound`. No project-specific counting axiom is added; the trusted basis includes the Lean kernel and the locked dependencies.

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